



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A++' Grade

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Class : CSE
Batch : 2025-29

Assignment : 23

Subject: Programming with Java

ASSIGNMENT No : 23

TITLE : Develop a Java application using JDBC to navigate and retrieve records from a ResultSet.

Problem Statement

Develop a Java application using JDBC to navigate and retrieve records from a ResultSet.

Learning Objectives

To navigate and retrieve records using ResultSet methods.

Theory

A ResultSet object contains data retrieved from a database after executing a SELECT query. It acts like a cursor that points to one row of the result at a time. Initially, the cursor is positioned before the first row. Various methods such as next(), previous(), first(), last(), absolute(), and relative() allow movement through the records. Data from each column can be retrieved using methods like getString(), getInt(), and getDouble(). Scrollable and updatable ResultSets provide greater flexibility for navigating and modifying retrieved data. ResultSet plays a crucial role in displaying and processing database information in Java applications.



Code :

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import java.sql.*;

public class Exp23 extends JFrame {
    private static final String DB_URL = "jdbc:mysql://localhost:3306/hospital_db";
    private static final String DB_USER = "root";
    private static final String DB_PASSWORD = "Abcd@1234";

    public Exp23() {
        setTitle("Hospital Portal System");
        setSize(450, 350);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setLocationRelativeTo(null);

        JTabbedPane tabbedPane = new JTabbedPane();
        tabbedPane.addTab("General Login", createGeneralLoginPanel());
        tabbedPane.addTab("Staff Login", createStaffLoginPanel());

        add(tabbedPane);
    }

    private JPanel createGeneralLoginPanel() {
        JPanel panel = new JPanel(new GridBagLayout());
        GridBagConstraints gbc = new GridBagConstraints();
        gbc.insets = new Insets(8, 8, 8, 8);
        gbc.fill = GridBagConstraints.HORIZONTAL;

        JLabel lblTitle = new JLabel("=== System Login ===", SwingConstants.CENTER);
        lblTitle.setFont(new Font("Arial", Font.BOLD, 16));

        JLabel lblUser = new JLabel("Username:");
        JTextField txtUser = new JTextField(15);
        JLabel lblPass = new JLabel("Password:");
        JPasswordField txtPass = new JPasswordField(15);
        JButton btnLogin = new JButton("Login");
```



```
JLabel lblStatus = new JLabel("", SwingConstants.CENTER);

gbc.gridx = 0; gbc.gridy = 0; gbc.gridwidth = 2;
panel.add(lblTitle, gbc);

gbc.gridwidth = 1;
gbc.gridx = 0; gbc.gridy = 1;
panel.add(lblUser, gbc);
gbc.gridx = 1; gbc.gridy = 1;
panel.add(txtUser, gbc);

gbc.gridx = 0; gbc.gridy = 2;
panel.add(lblPass, gbc);
gbc.gridx = 1; gbc.gridy = 2;
panel.add(txtPass, gbc);

gbc.gridx = 0; gbc.gridy = 3; gbc.gridwidth = 2;
panel.add(btnLogin, gbc);

gbc.gridx = 0; gbc.gridy = 4; gbc.gridwidth = 2;
panel.add(lblStatus, gbc);

btnLogin.addActionListener(new ActionListener() {
    @Override
    public void actionPerformed(ActionEvent e) {
        String username = txtUser.getText();
        String password = new String(txtPass.getPassword());

        String query = "SELECT * FROM users WHERE username = ? AND password =
?";

        try (Connection conn = DriverManager.getConnection(DB_URL, DB_USER,
DB_PASSWORD);
            PreparedStatement pstmt = conn.prepareStatement(query)) {

            pstmt.setString(1, username);
            pstmt.setString(2, password);

            try (ResultSet rs = pstmt.executeQuery()) {
                if (rs.next()) {
```



```
        lblStatus.setForeground(Color.GREEN.darker());
        lblStatus.setText("Login Successful! Welcome, " + rs.getString("username")
+ ".");
    } else {
        lblStatus.setForeground(Color.RED);
        lblStatus.setText("Invalid Username or Password.");
    }
}
} catch (Exception ex) {
    lblStatus.setForeground(Color.RED);
    lblStatus.setText("Database Error: " + ex.getMessage());
}
}
});

return panel;
}

private JPanel createStaffLoginPanel() {
    JPanel panel = new JPanel(new GridBagLayout());
    GridBagConstraints gbc = new GridBagConstraints();
    gbc.insets = new Insets(8, 8, 8, 8);
    gbc.fill = GridBagConstraints.HORIZONTAL;

    JLabel lblTitle = new JLabel("=== Hospital Staff Portal ===",
SwingConstants.CENTER);
    lblTitle.setFont(new Font("Arial", Font.BOLD, 16));

    JLabel lblStaffId = new JLabel("Staff ID:");
    JTextField txtStaffId = new JTextField(15);
    JLabel lblPass = new JLabel("Password:");
    JPasswordField txtPass = new JPasswordField(15);
    JButton btnLogin = new JButton("Login");
    JLabel lblStatus = new JLabel("", SwingConstants.CENTER);

    gbc.gridx = 0; gbc.gridy = 0; gbc.gridwidth = 2;
    panel.add(lblTitle, gbc);

    gbc.gridwidth = 1;
```



```
gbc.gridx = 0; gbc.gridy = 1;
panel.add(lblStaffId, gbc);
gbc.gridx = 1; gbc.gridy = 1;
panel.add(txtStaffId, gbc);
```

```
gbc.gridx = 0; gbc.gridy = 2;
panel.add(lblPass, gbc);
gbc.gridx = 1; gbc.gridy = 2;
panel.add(txtPass, gbc);
```

```
gbc.gridx = 0; gbc.gridy = 3; gbc.gridwidth = 2;
panel.add(btnLogin, gbc);
```

```
gbc.gridx = 0; gbc.gridy = 4; gbc.gridwidth = 2;
panel.add(lblStatus, gbc);
```

```
btnLogin.addActionListener(new ActionListener() {
    @Override
    public void actionPerformed(ActionEvent e) {
        String staffId = txtStaffId.getText();
        String password = new String(txtPass.getPassword());
```

```
        String query = "SELECT role FROM hospital_staff WHERE staff_id = ? AND
password = ?";
```

```
        try (Connection conn = DriverManager.getConnection(DB_URL, DB_USER,
DB_PASSWORD);
```

```
            PreparedStatement pstmt = conn.prepareStatement(query)) {
```

```
            pstmt.setString(1, staffId);
            pstmt.setString(2, password);
```

```
            try (ResultSet rs = pstmt.executeQuery()) {
                if (rs.next()) {
                    String role = rs.getString("role");
                    lblStatus.setForeground(Color.GREEN.darker());
```

```
                    if ("Doctor".equalsIgnoreCase(role)) {
                        lblStatus.setText("Access Granted: Welcome, Dr. " + staffId);
                    } else if ("Nurse".equalsIgnoreCase(role)) {
```



```
        lblStatus.setText("Access Granted: Welcome, Nurse " + staffId);
    } else {
        lblStatus.setText("Access Granted: Welcome, Staff member.");
    }
} else {
    lblStatus.setForeground(Color.RED);
    lblStatus.setText("Authentication Failed: Invalid Staff ID or Password.");
}
}
} catch (Exception ex) {
    lblStatus.setForeground(Color.RED);
    lblStatus.setText("Database Error: " + ex.getMessage());
}
}
});

return panel;
}

public static void main(String[] args) {
    SwingUtilities.invokeLater(() -> {
        new Exp23().setVisible(true);
    });
}
}
```



SQL Pre-Requisites :

```
1 • CREATE DATABASE hospital_db;
2 • USE hospital_db;
3 • CREATE TABLE hospital_staff (
4     id INT PRIMARY KEY AUTO_INCREMENT,
5     staff_id VARCHAR(50) UNIQUE NOT NULL,
6     password VARCHAR(50) NOT NULL,
7     role VARCHAR(20) NOT NULL
8 );
9 • INSERT INTO hospital_staff (staff_id, password, role) VALUES
10 ('DOC205', 'medpass', 'Doctor'),
11 ('NUR310', 'nursepass', 'Nurse');
12
13 • CREATE TABLE users (
14     id INT PRIMARY KEY AUTO_INCREMENT,
15     username VARCHAR(50) NOT NULL,
16     password VARCHAR(50) NOT NULL
17 );
18 • INSERT INTO users (username, password) VALUES ('doctor1', 'pass123');
19
```

Output :

Valid General Login :

Hospital Portal System

General Login | Staff Login

=== System Login ===

Username:

Password:

Login Successful! Welcome, doctor1.



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Invalid Login :

The screenshot shows a web browser window titled "Hospital Portal System". It has two tabs: "General Login" and "Staff Login". The "General Login" tab is active. The page displays the heading "=== System Login ===". Below this, there are two input fields: "Username:" with the value "doctor2" and "Password:" with masked characters ".....". A "Login" button is positioned below the password field. At the bottom of the form, a red error message reads "Invalid Username or Password."

Valid Staff Login :

The screenshot shows the same "Hospital Portal System" web browser window, but the "Staff Login" tab is now active. The page displays the heading "=== Hospital Staff Portal ===". Below this, there are two input fields: "Staff ID:" with the value "DOC205" and "Password:" with masked characters ".....". A "Login" button is positioned below the password field. At the bottom of the form, a green success message reads "Access Granted: Welcome, Dr. DOC205".



Conclusion :

The Hospital Portal System was successfully implemented using Java Swing and MySQL. The application provides separate General Login and Hospital Staff Login interfaces. JDBC was used to establish a connection between the Java application and the MySQL database, while PreparedStatement was used to securely execute login queries. The system successfully authenticates users and provides role-based access for doctors and nurses.

Exercise

Q1) Create a program to display database records one by one using ResultSet.

Sol) Code :

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;

public class Disprec {

    public static void main(String[] args) {

        String url = "jdbc:mysql://localhost:3306/hospital_db";
        String user = "root";
        String password = "Abcd@1234";

        String query = "SELECT staff_id, role FROM hospital_staff";

        try {
            Class.forName("com.mysql.cj.jdbc.Driver");

            Connection con = DriverManager.getConnection(url, user, password);

            PreparedStatement ps = con.prepareStatement(query);

            ResultSet rs = ps.executeQuery();

            System.out.println("==== Hospital Staff Records ====");

            while (rs.next()) {
                System.out.println("Staff ID : " + rs.getString("staff_id"));
                System.out.println("Role    : " + rs.getString("role"));
            }
        }
    }
}
```



```
        System.out.println("-----");
    }

    rs.close();
    ps.close();
    con.close();

} catch (Exception e) {
    System.out.println("Database Error: " + e.getMessage());
}
}
```

SQL Table :

```
3 • CREATE TABLE hospital_staff (
4     id INT PRIMARY KEY AUTO_INCREMENT,
5     staff_id VARCHAR(50) UNIQUE NOT NULL,
6     password VARCHAR(50) NOT NULL,
7     role VARCHAR(20) NOT NULL
8 );
9 • INSERT INTO hospital_staff (staff_id, password, role) VALUES
10 ('DOC205', 'medpass', 'Doctor'),
11 ('NUR310', 'nursepass', 'Nurse');
12
13 • INSERT INTO hospital_staff (staff_id, password, role) VALUES
14 ('DOC101', 'docpass', 'Doctor'),
15 ('DOC102', 'doctor123', 'Doctor'),
16 ('NUR301', 'nurse123', 'Nurse'),
17 ('NUR302', 'nurse456', 'Nurse') ;
18
```



Output :

```
Problems 29 Output Debug Console Terminal Ports
● PS C:\Users\LENOVO\Desktop\JDBC\src> javac -cp "../lib/*" Disprec.java
● PS C:\Users\LENOVO\Desktop\JDBC\src> java -cp ".;../lib/*" Disprec
=== Hospital Staff Records ===
Staff ID : DOC205
Role      : Doctor
-----
Staff ID : NUR310
Role      : Nurse
-----
Staff ID : DOC101
Role      : Doctor
-----
Staff ID : DOC102
Role      : Doctor
-----
Staff ID : NUR301
Role      : Nurse
-----
Staff ID : NUR302
Role      : Nurse
-----
PS C:\Users\LENOVO\Desktop\JDBC\src>
```

Q2) Create a Java program that retrieves employee records from a database and displays them one by one using ResultSet. Show employee ID, name, department, and salary sequentially.

Sol) Code :

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
```

```
public class Dispemp {
```

```
    public static void main(String[] args) {
```

```
        String url = "jdbc:mysql://localhost:3306/hospital_db";
        String user = "root";
        String password = "Abcd@1234";
```



```
String query = "SELECT emp_id, name, department, salary FROM employee";
```

```
try {
```

```
    Class.forName("com.mysql.cj.jdbc.Driver");
```

```
    Connection con = DriverManager.getConnection(url, user, password);
```

```
    PreparedStatement ps = con.prepareStatement(query);
```

```
    ResultSet rs = ps.executeQuery();
```

```
    System.out.println("=== Employee Records ===");
```

```
    while (rs.next()) {
```

```
        System.out.println("Employee ID : " + rs.getInt("emp_id"));
```

```
        System.out.println("Name      : " + rs.getString("name"));
```

```
        System.out.println("Department : " + rs.getString("department"));
```

```
        System.out.println("Salary    : " + rs.getDouble("salary"));
```

```
        System.out.println("-----");
```

```
    }
```

```
    rs.close();
```

```
    ps.close();
```

```
    con.close();
```

```
} catch (Exception e) {
```

```
    System.out.println("Database Error: " + e.getMessage());
```

```
}
```

```
}
```

```
}
```



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SQL Table :

```
30 • CREATE TABLE employee (  
31     emp_id INT,  
32     name VARCHAR(50),  
33     department VARCHAR(50),  
34     salary DOUBLE  
35 );  
36  
37 • INSERT INTO employee VALUES  
38 (101, 'Rahul', 'IT', 55000),  
39 (102, 'Priya', 'HR', 48000),  
40 (103, 'Aman', 'Finance', 60000),  
41 (104, 'Sneha', 'IT', 58000),  
42 (105, 'Rohan', 'Marketing', 45000);  
43
```

Output :

```
Problems 32 Output Debug Console Terminal Ports  
• PS C:\Users\LENOVO\Desktop\JDBC\src> javac -cp "../lib/*" Dispemp.java  
• PS C:\Users\LENOVO\Desktop\JDBC\src> java -cp ".;../lib/*" Dispemp  
=== Employee Records ===  
Employee ID : 101  
Name       : Rahul  
Department : IT  
Salary      : 55000.0  
-----  
Employee ID : 102  
Name       : Priya  
Department : HR  
Salary      : 48000.0  
-----  
Employee ID : 103  
Name       : Aman  
Department : Finance  
Salary      : 60000.0  
-----  
Employee ID : 104  
Name       : Sneha  
Department : IT  
Salary      : 58000.0  
-----  
Employee ID : 105  
Name       : Rohan  
Department : Marketing  
Salary      : 45000.0  
-----  
❖ PS C:\Users\LENOVO\Desktop\JDBC\src> |
```